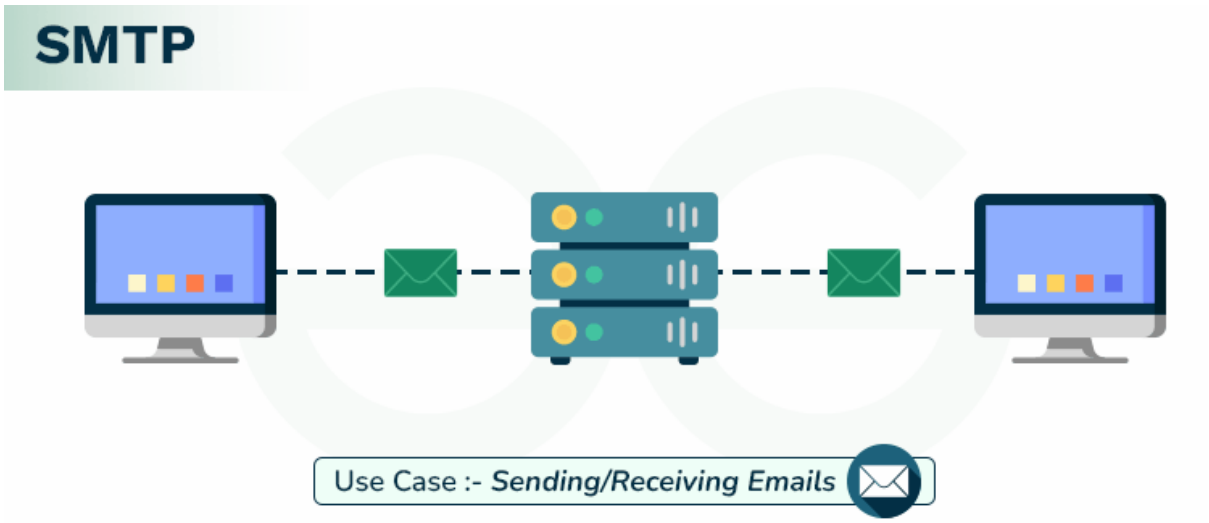
Simple Mail Transfer Protocol (SMTP)

Simple Mail Transfer Protocol (SMTP) is an application layer protocol used for exchanging email messages between servers. It is essential in the email communication process and operates at the application layer of the TCP/IP stack.
To send an email, the client opens a TCP connection to the SMTP server. The server, which is always listening on port 25, initiates the connection as soon as it detects a client. Once the TCP connection is established, the client sends the email across the connection.



Features

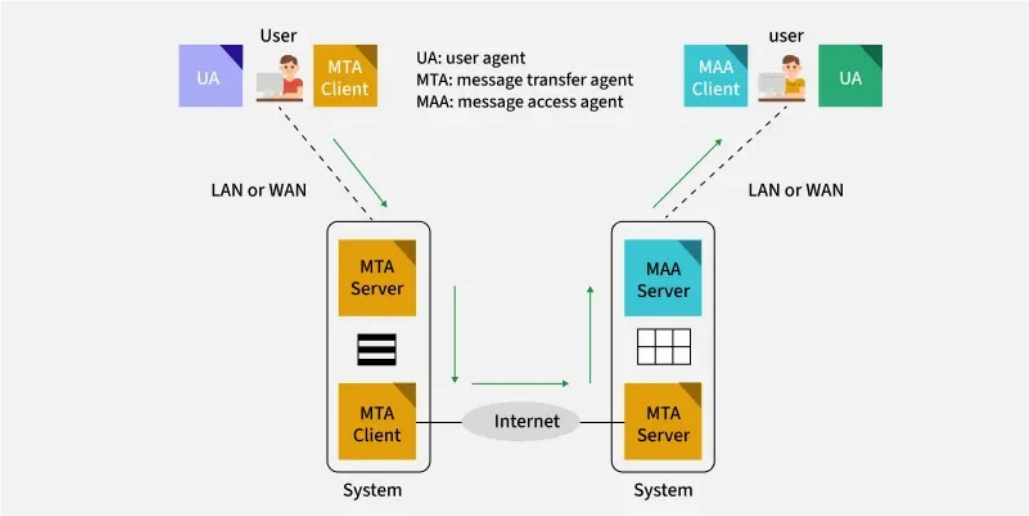
- **Dedicated IP**
- Email API: Integrating SMTP.com with your business can be easy with the email API feature. They have complete API documentation on their website that can help you integrate your business in just 5 minutes.
- 24x7 Customer Support: The round-the-clock support is one of the best features of SMTP.com. Support is available both on the website and also for paid customers. 24x7, all human support is available for all customers across all plans. No third party is involved and solutions are provided fast for easy implementation. Online chat support is also available for those who are looking for more information about SMTP.com
- High Volume Sending Solutions: This newly launched feature is great for those businesses who want to send more than 250 million emails a month. Customized quotations and solutions are available.
- Reputation Defender: This is an add-on feature that helps clean up your email lists. It doesn't need any integration but actively monitors your lists and provides a report.

Components of SMTP

- **Mail User Agent (MUA):** It is a computer application that helps you in sending and retrieving mail. It is responsible for creating email messages for transfer to the mail transfer agent(MTA).

- **Mail Submission Agent (MSA):** It is a computer program that receives mail from a Mail User Agent(MUA) and interacts with the Mail Transfer Agent(MTA) for the transfer of the mail.
- **Mail Transfer Agent (MTA):** It is software that has the work to transfer mail from one system to another with the help of SMTP.
- **Mail Delivery Agent (MDA):** A mail Delivery agent or Local Delivery Agent is basically a system that helps in the delivery of mail to the local system.

How does SMTP Work?



SMTP

1. Sending Email:
- When a user wants to send an email, they use a **User Agent (UA)**, like Outlook or Gmail.
- The email is handed over to the **MTA**, which is responsible for transferring the email to the recipient's mail server.

2. SMTP Client and Server:
- **Sender-SMTP (Client)**: The email sender's MTA initiates the connection to the recipient's MTA (Receiver-SMTP).
- **Receiver-SMTP (Server)**: The receiving MTA listens for incoming connections and receives the email from the sender-SMTP.
- This communication happens over **TCP port 25**.

3. Relays and Gateways:
- **Relays**: In some cases, the email may pass through several intermediate MTAs before reaching the destination server. These MTAs act as **relays**.
- **Gateways**: If the sending and receiving systems use different email protocols (e.g., SMTP and non-SMTP), an **email gateway** can convert the email to the appropriate format for delivery.

4. Email Delivery:
- The sender's **MTA** sends the email to the **receiver's MTA**, either directly or through relays.
- The **MTA** uses the **SMTP** protocol to transfer the message. Once it's delivered to the destination MTA, the email is placed in the recipient's mailbox.
- The recipient's **User Agent (UA)** can then download the email.